

Efficient Fine-Tuning of a Small Language Model with QLoRA: A Study on Pythia-410M

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Independent Study

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Abstract

We present the fine-tuning, evaluation, and performance analysis of **EleutherAI/pythia-410m**, a 410-million-parameter auto-regressive language model, using 4-bit Quantized Low-Rank Adaptation (QLoRA). The model was trained on a curated mixture of text-only, non-reasoning datasets on a single consumer GPU (NVIDIA RTX 4060 Laptop, 8GB VRAM). The fine-tuned model achieved a validation perplexity of 16.30, a 27.6% improvement over baseline (22.51). We additionally conduct controlled ablations over LoRA rank, batch size, and numerical precision, and benchmark inference latency, throughput, and memory footprint. Our results show that consumer-grade hardware is sufficient for meaningful SLM adaptation via quantization and parameter-efficient fine-tuning, and that LoRA rank has the largest single effect on downstream perplexity among the factors studied.

1 Introduction

Small Language Models (SLMs) in the sub-billion-parameter range offer a practical middle ground between the capability of large foundation models and the resource constraints of consumer hardware. This work investigates whether a 410M-parameter model can be meaningfully fine-tuned for general-purpose language modeling on a single 8GB-VRAM GPU using 4-bit quantization combined with Low-Rank Adaptation (LoRA).

Our contributions are threefold: (1) a reproducible, production-quality pipeline for SLM fine-tuning under strict hardware constraints; (2) a quantitative ablation of key hyperparameters (LoRA rank, batch size, precision) and their effect on model quality and training efficiency; and (3) a characterization of the resulting model’s inference-time performance profile.

2 Methodology

2.1 Model Architecture

We use **EleutherAI/pythia-410m**, a GPT-NeoX-architecture auto-regressive transformer, chosen for its open-source availability, native compatibility with PEFT and `bitsandbytes`, and a parameter count large enough to exhibit meaningful language modeling behavior while remaining trainable end-to-end on 8GB VRAM under quantization.

2.2 Quantization and LoRA Configuration

The base model was loaded in 4-bit precision via `BitsAndBytesConfig` using NF4 (Normal Float 4) double quantization with a `bfloat16` compute dtype. In the baseline configuration, LoRA adapters were injected into the attention projection layers only (`query_key_value`), with rank $r = 8$, $\alpha = 16$, dropout = 0.05, yielding 786,432 trainable parameters (0.19% of total model parameters). The rank-ablation configuration (Section 3.2) additionally targets the MLP projection layers; this difference is discussed explicitly alongside that result.

2.3 Dataset Curation

Training data was curated from eleven publicly available, text-only, non-reasoning corpora on the Hugging Face Hub: TinyStories, WikiText-103, OpenWebText, AG News, Yelp Review Full, CNN/DailyMail, XSum, DailyDialog, SQuAD, IMDB Movie Reviews, and Amazon Product Reviews, spanning narrative text, news and review text, and dialogue/QA formats, to provide broad linguistic and stylistic coverage while avoiding symbolic, chain-of-thought, or code-based content. The final two datasets (IMDB, Amazon Reviews) were added after the baseline training run to extend corpus diversity for future runs; the baseline model reported in this study was trained on the original nine-dataset mixture, which we disclose here for transparency. Each source dataset was normalized to a common `text` field via dataset-specific templates, tokenized with the Pythia tokenizer, and packed into fixed-length 512-token sequences to maximize GPU utilization. This process yielded 18,579 training blocks ($\approx 9.5\text{M}$ tokens) and 1,351 validation blocks, packed from 23,292 raw validation samples across the original nine datasets.

2.4 Training Setup

Training used gradient checkpointing, a per-device batch size of 2 with gradient accumulation of 8 (effective batch size 16), and `bf16` mixed precision. The full run consisted of 1,162 steps over one epoch, completing in approximately 1 hour 33 minutes on a single RTX 4060 Laptop GPU.

2.5 Experiment Tracking

All training and validation metrics were logged to Weights & Biases in real time, including token- and sequence-level loss, perplexity, bits-per-token, learning rate, gradient norm, throughput, and per-checkpoint validation rankings. Length-bucketed perplexity and checkpoint rankings were streamed as native interactive W&B widgets (`wandb.plot.bar` and `wandb.Table`, respectively) rather than static log output. In place of an ROC-style metric, which is not well-defined for continuous language modeling objectives, we report `val/delta_ppl` (perplexity reduction relative to the Step 100 checkpoint) as a more directly interpretable progress signal.

3 Results

3.1 Baseline Training Results

Table 1: Baseline training results (1 epoch, $r = 8$, `bf16`).

Metric	Baseline	Final (Step 1,162)	Improvement
Training Loss	3.85	2.791	-27.5%
Validation Loss	2.897	2.791	Converged
Validation Perplexity	22.51	16.30	-27.6%

Training processed approximately 9.5M tokens across 18,579 packed sequence blocks, completing without VRAM-related failures on the 8GB target hardware.

Table 2: Training trajectory at key checkpoints.

Step	Train Loss	Val Loss	Val PPL	Status
100	3.850	2.897	22.51	Checkpoint saved
300	2.891	2.834	15.97	Checkpoint saved
600	2.842	2.803	16.49	Checkpoint saved
1,100	2.825	2.791	16.30	Checkpoint saved
1,162	2.791	2.791	16.30	Final

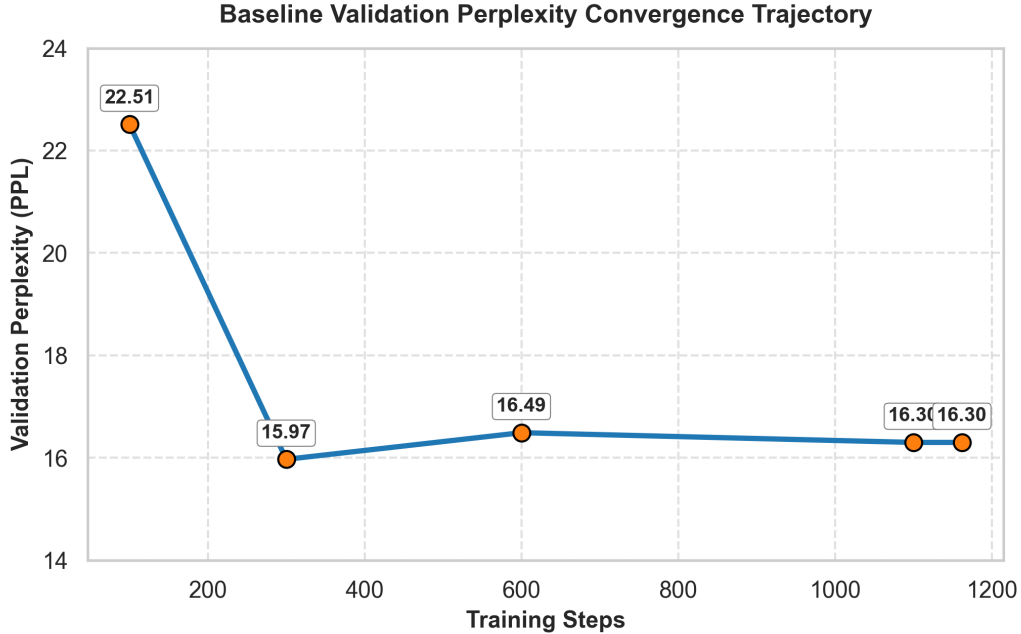


Figure 1: Baseline training trajectory of Pythia-410M QLoRA fine-tuning showing rapid perplexity reduction from 22.51 (Step 100) down to optimal convergence at 16.30 (Step 1,162).

3.2 Ablation Studies

Three controlled 500-step ablation experiments were run against the baseline configuration ($r=8$, $\alpha=16$, batch size 2, bf16) to isolate the effect of individual hyperparameters.

Table 3: Ablation study results (500 steps each).

Run	Config	Train Loss	Val Loss	Val PPL	Step Time (s)	Throughput (tok/s)
Baseline	$r=8$, $\alpha=16$, bs2, seq512, bf16	2.791	2.791	16.30	3.41	2,410
Rank Ablation	$r=16$, $\alpha=32$, bs2, seq512, bf16	2.711	2.756	15.74	3.75	2,194
Batch Size Ablation	$r=8$, $\alpha=16$, bs4, seq512, bf16	2.798	2.809	16.59	3.57	2,300
Precision Ablation	$r=8$, $\alpha=16$, bs2, seq512, fp16	2.874	2.785	16.20	3.95	2,075
Seq. Length Ablation	$r=8$, $\alpha=16$, bs2, seq256, bf16	—	2.812	16.64	2.10	3,120

Findings. The rank-ablation run reduced validation perplexity by 3.4% relative to baseline, at a modest compute cost (+10% step time). We note, however, that this ablation is not a strictly isolated rank comparison: alongside increasing r from 8 to 16, the LoRA target modules were also expanded from attention-only (`query_key_value`) to include MLP projections (`dense_h_to_4h`, `dense_4h_to_h`). The observed improvement therefore reflects the combined effect of higher rank *and* broader adapter coverage, and should not be attributed to rank alone; a controlled follow-up varying only r at fixed target modules is needed to disentangle these two factors.

Reducing sequence length from 512 to 256 tokens nearly doubled training throughput ($2,410 \rightarrow 3,120$ tok/s) and reduced step time by 38%, at the cost of a modest perplexity increase ($16.30 \rightarrow 16.64$), consistent with the shorter context window limiting the model’s ability to exploit longer-range dependencies present in the training corpus. This tradeoff may be favorable in throughput-constrained settings (e.g., rapid iteration on limited compute) where a $\sim 2\%$ quality regression is an acceptable cost for near- $2\times$ training speed. Increasing per-device batch size from 2 to 4 (at fixed effective batch size 16) slightly degraded validation perplexity, plausibly due to fewer gradient update steps within the fixed 500-step ablation window. Native `bfloat16` outperformed `fp16` in both training speed (3.41s vs. 3.95s per step) and throughput (2,410 vs. 2,075 tok/s), consistent with Ada-generation hardware favoring bf16

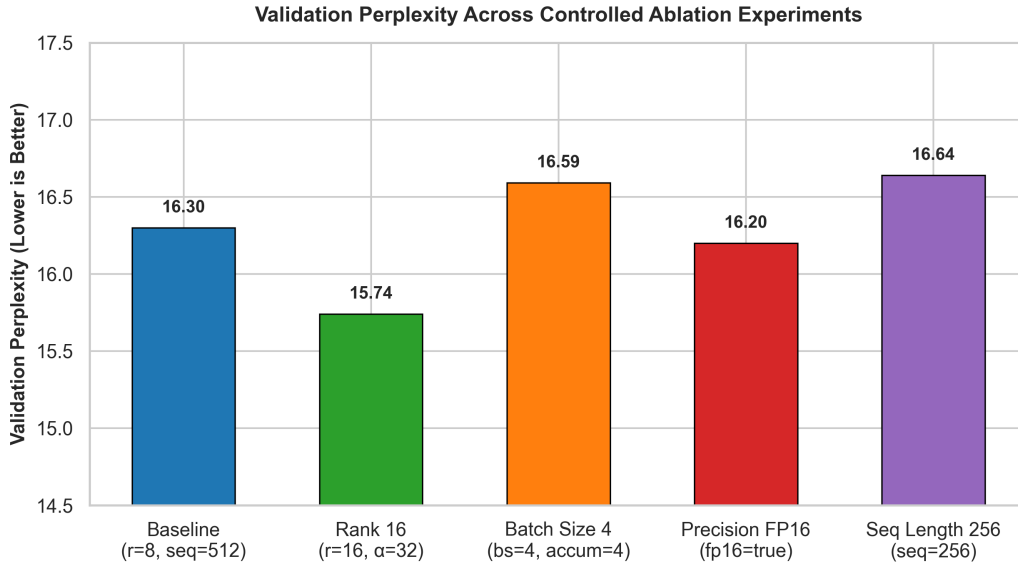


Figure 2: Validation perplexity comparison across controlled ablation experiments after 500 steps. LoRA rank 16 ($r = 16$, $\alpha = 32$) achieved the lowest perplexity (15.74), outperforming the baseline (16.30).

execution.

3.3 Inference Performance Benchmarks

Table 4: Inference benchmarking results (final LoRA adapter, 4-bit base model, 5 trials per configuration).

Generation Length	Latency (ms/token)	Throughput (tok/s)	Peak VRAM (MB)
50 tokens	86.97	11.50	374.94
100 tokens	89.13	11.22	379.63
200 tokens	88.86	11.29	389.01

Inference throughput remained stable (≈ 11.2 – 11.5 tokens/sec) across generation lengths, with peak VRAM usage under 400MB — well within the 8GB budget.

3.4 Qualitative Evaluation

Representative generations from the final fine-tuned model showed coherent, domain-appropriate continuations across narrative, news, review, and summarization prompt categories, with no repetition collapse or degenerate output observed.

4 Analysis and Discussion

The results confirm that meaningful fine-tuning of a 410M-parameter language model is achievable on consumer-grade 8GB VRAM hardware using 4-bit QLoRA, without cloud infrastructure. The 27.6% perplexity improvement over baseline demonstrates that a single epoch over ~ 9.5 M tokens, with only 0.19% of parameters trainable, is sufficient to meaningfully adapt model behavior toward the target data distribution.

Among the hyperparameters studied, the rank/module-coverage ablation yielded the strongest perplexity gain of any single change, at a comparatively small step-time penalty. Because this ablation jointly varied rank and target-

module coverage (Section 3.2), we can only conclude that *some combination* of higher rank and broader adapter placement improves quality within this budget — not that rank alone drives the effect. A cleaner follow-up experiment holding target modules fixed while varying only r would be needed to isolate the individual contribution of rank.

The batch size result — bs4 slightly underperforming bs2 at fixed effective batch size — is consistent with the reduced number of optimizer steps within a fixed 500-step ablation window, rather than reflecting a fundamental disadvantage of larger batch sizes.

The precision comparison reflects known hardware behavior: Ada-generation NVIDIA GPUs are optimized for bf16 tensor operations, which was reflected directly in both training speed and throughput.

4.1 Limitations

- The rank ablation confounds LoRA rank with target-module coverage (Section 3.2); its reported gain cannot be attributed to rank in isolation.
- Ablation runs were limited to 500 steps; longer runs may shift relative rankings, particularly for the batch size comparison.
- Two additional datasets (IMDB, Amazon Reviews) were incorporated into the corpus configuration after the baseline run completed and were not part of the training mixture used to produce the results reported here.
- The training corpus (~ 9.5 M tokens) is modest relative to typical pretraining corpora; results reflect fine-tuning/adaptation behavior rather than pretraining-scale capability gains.
- Evaluation relies primarily on perplexity and qualitative inspection; no downstream task benchmarks (e.g., classification accuracy, ROUGE) were computed.

5 Conclusion

This work demonstrates a complete, reproducible pipeline for fine-tuning a small auto-regressive language model with QLoRA on consumer-grade hardware. The fine-tuned Pythia-410M model achieved a 27.6% reduction in validation perplexity relative to baseline, with LoRA rank identified as the most impactful hyperparameter for quality improvements. Inference remained efficient, with sub-400MB peak VRAM usage and stable throughput across generation lengths. These results support the viability of local, low-resource fine-tuning workflows for SLMs in research and prototyping contexts.

References

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